

Transformations & Constructions

Test · 27 marks total

Name: _____ Date: _____ Score: _____ / 27

Section A — Translation

A1. Translate the point A(2, 5) by the vector (3, -4). State the coordinates of the image.

[2 marks]

A2. Find the column vector that translates the point (1, 1) to the point (4, -2).

[2 marks]

A3. True or false: a translation changes the size of a shape.

[1 mark]

Section B — Rotation

B1. Name the three pieces of information needed to fully describe a rotation.

[3 marks]

B2. Rotate the point (2, 3) by 180° about the origin. State the coordinates of the image.

[2 marks]

Section C — Reflection

C1. Reflect the point (3, 2) in the y-axis. State the coordinates of the image.

[1 mark]

C2. Reflect the point (1, 4) in the line $y = x$. State the coordinates of the image.

[1 mark]

C3. The point P(5, 1) is reflected in the line $x = 2$. Find the coordinates of the image.

[2 marks]

Section D — Enlargement

D1. A shape is enlarged by scale factor 3 from a centre of enlargement. A side on the object is 4 cm. Find the length of the corresponding side on the image.

[1 mark]

D2. A shape is enlarged by scale factor $\frac{1}{2}$. What effect does this have on the size of the shape?

[1 mark]

D3. A triangle has area 8 cm^2 . It is enlarged by scale factor 3. Find the area of the enlarged triangle.

[2 marks]

D4. A shape is enlarged by scale factor -2 about a centre of enlargement. Describe two effects this has on the image compared with the object.

[2 marks]

D5. Explain why a scale factor between 0 and 1 still counts as an "enlargement", even though the shape gets smaller.

[1 mark]

Section E — Constructions

E1. Describe how to construct the perpendicular bisector of a line segment AB, using only a ruler and compasses.

[3 marks]

E2. Describe how to construct the bisector of an angle, using only a ruler and compasses.

[2 marks]

E3. Why must construction lines and arcs never be rubbed out on an exam answer?

[1 mark]